Hon Ching Li

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Active DoD Secret Clearance

Education

University at Buffalo, The State University of New York **02/2022 - 02/2024**
Master of Science in Computer Science, GPA 3.67/4.0 *Buffalo, NY*
Relevant Courses: Algorithms Analysis and Design I (A), Software Engineering (A-), Software Security (A)
Computer Architecture (A), Machine Learning (A-), Reinforcement Learning (A)

University at Buffalo, The State University of New York **08/2016 - 02/2022**
Bachelor of Science in Computer Science, GPA 3.33/4.0 *Buffalo, NY*
Relevant Courses: College Calculus 1 (A), College Calculus 2 (A), Linear Algebra (A-)
Intro to Probability (A), Machine Learning (A-), Data Mining (A-)

Technical Skills

Programming Languages: Python, C++, R, SQL, Java
Quantitative Finance & Risk Modeling: Algorithmic Backtesting, Portfolio Optimization, Monte Carlo Simulation, Time-Series Analysis, Value at Risk (VaR), Conditional Value at Risk (CVaR), Sharpe Ratio, Maximum Drawdown
Statistical Computing & Machine Learning: NumPy, pandas, SciPy, scikit-learn, PyTorch, XGBoost, Regression Analysis, Reinforcement Learning (DQN/DDQN)
Optimization & Numerical Methods: Linear Programming, Quadratic Programming, Mixed-Integer Optimization, Stochastic Simulation, Scenario Analysis, Sensitivity Analysis
Development Tools & Infrastructure: Jupyter Notebook, Matplotlib, Flask, PostgreSQL, Docker, AWS, Git, Linux

Work Experience

25H Network Communication Systems Specialist, U.S. Army Reserve **06/2025 - Present**
Army Reserve Soldier - Radio and Transport Communications

- Operate and maintain **radio, transport, and communications network systems** supporting voice and data services, including equipment setup, connectivity verification, and service continuity.
- Apply **routing and switching** fundamentals, **IP addressing**, and structured troubleshooting to assist with network changes, fault isolation, and service restoration.
- Use technical manuals, signal-flow diagrams, and diagnostic procedures to identify issues, isolate faults, and restore communications availability.
- Follow **COMSEC/OPSEC** and **information assurance** procedures when handling sensitive communications equipment, documentation, and mission-related information.

FLAZ Lab, Inc. **05/2026 - Present**
Co-founder

- Developed a **Python-based analytics pipeline** to process 3,600+ e-commerce transactions across 18 products, analyzing sales patterns, pricing sensitivity, and demand variability to support data-driven business decisions.
- Built SKU-level demand forecasting models using **XGBoost** and time-series features, reducing out-of-sample forecast error by 25% compared with a historical-average baseline.
- Formulated a **mixed-integer optimization model** to allocate a $15K inventory budget under demand uncertainty, supplier lead times, and purchasing constraints; achieved 12% higher simulated contribution profit and 17% lower excess inventory in historical backtests.

First Quality Pharmacy, Inc. **10/2024 - Present**
Co-Founder

- Co-founded an independent pharmacy, contributing to strategic planning, **inventory management**, and financial decision-making in a competitive healthcare market.
- Analyzed demand fluctuations, supplier pricing, and operating costs through **scenario and sensitivity analyses** to evaluate downside exposure and inform purchasing decisions.
- Assessed **inventory and liquidity risks** by monitoring demand variability, product expiration, and cash flow requirements to guide resource allocation and operational planning.

Teaching Assistant, University at Buffalo **01/2023 - 05/2023**
Computer Architecture Course (Graduate Level, CPU Design) *Buffalo, NY*

- Co-led a team of five teaching assistants supporting 200+ students in a graduate computer architecture course. The course average rose by more than 18% compared with previous terms.
- Created and taught a **Verilog** tutorial on single-cycle CPU design and graded quizzes and exams.

Projects

Neutral Grid Backtesting Service | *Python, Flask, Docker*

- Developed a **deterministic, state-driven Python backtesting engine** for a neutral-grid trading strategy, simulating order execution, position management, and portfolio equity over 24 months of historical market data.
- Designed **risk-adjusted performance analytics** to evaluate cumulative net returns, annualized **Sharpe Ratio**, and **Maximum Drawdown**. Illustrative results: 21.6% cumulative net return, 1.18 Sharpe Ratio, and 13.4% Maximum Drawdown after estimated transaction costs.
- Conducted **parameter sensitivity analysis** across 72 grid-spacing and position-sizing configurations, comparing risk-adjusted returns and downside exposure under different market conditions.

Atari Breakout Reinforcement Learning | *Python, PyTorch*

- Developed a Double Deep Q-Network (DDQN) agent using **Bellman-based dynamic programming** and temporal-difference learning to optimize sequential decisions and maximize long-term cumulative rewards.
- Implemented CNN-based **value function approximation**, Double Q-learning, and mini-batch experience replay to mitigate **Q-value overestimation bias** and improve training stability.
- Led a three-member team in training the agent over approximately 20,000 episodes, achieving an average Breakout score of **350 points**—approximately **11× the published human reference** of 31.8.

Pixel Art Converter (iOS) | *React Native, Node.js, Spring Boot, AWS, Docker*

- App Store: https://apps.apple.com/us/app/pixel-art-converter/id6742040975
- Engineered a **deterministic image-processing pipeline** using nearest-neighbor resampling and palette-based color transformations, supporting 50 pixelation levels and 20 color palettes.
- Optimized computational efficiency through **in-memory image processing**, minimizing disk I/O overhead and enabling real-time previews across configurable image transformations.
- Developed and deployed containerized **Node.js** image-processing and **Spring Boot** backend services on **AWS ECS**, supporting a production iOS application with 10,000+ downloads.

Certifications & Involvements

CompTIA Security+ ce — **2026**
U.S. DoD Security Clearance - Secret — **Active**
Build & Deploy AI Applications with Python & Flask — **IBM, 03/2026**
HTML, CSS, and JavaScript for Web Developers — **Johns Hopkins University**
Technical Director, UB SASE Executive Board — **2021 - 2023**